

Fuzzy Logic

Introduction

Example of Application areas of Fuzzy Logic:

- Fuzzy Control
 - Subway trains
 - Cement kilns
 - Washing Machines
 - Fridges

Fuzzy Sets

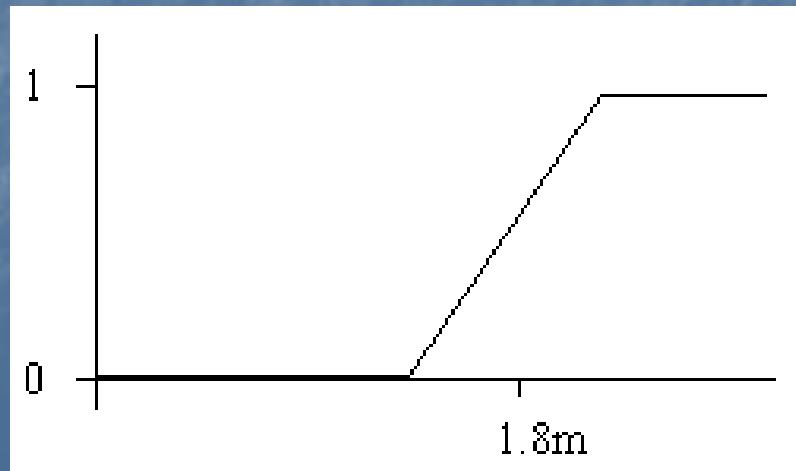
- Extension of Classical Sets
- Not just a membership value of in the set and out the set, 1 and 0
 - *but partial membership value, between 1 and 0*

Example: Height

- Tall people: say taller than or equal to 1.8m
 - 1.8m , 2m, 3m etc member of this set
 - 1.0 m, 1.5m or even 1.79999m not a member
- Real systems have measurement uncertainty
 - so near the border lines, many misclassifications

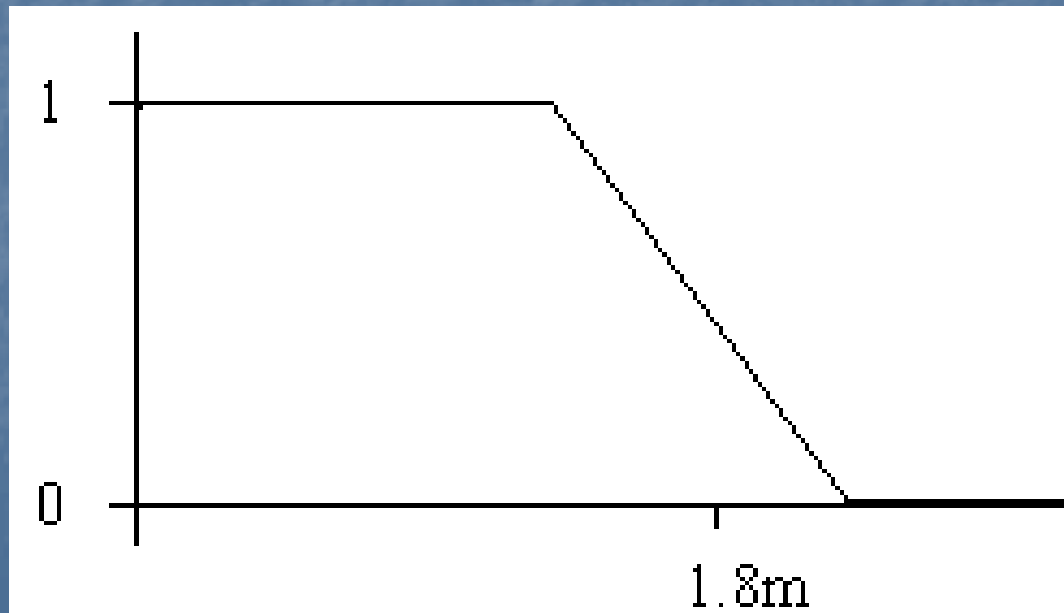
Member Functions

- Membership function
 - better than listing membership values
- e.g. $\text{Tall}(x) = \begin{cases} 1 & \text{if } x \geq 1.9\text{m} \\ 0 & \text{if } x \leq 1.7\text{m} \\ \text{else } (x - 1.7) / 0.2 \end{cases}$



Example: Fuzzy Short

- $\text{Short}(x) = \begin{cases} 0 & \text{if } x \geq 1.9\text{m} \\ 1 & \text{if } x \leq 1.7\text{m} \\ \text{else } (1.9 - x) / 0.2 \end{cases}$

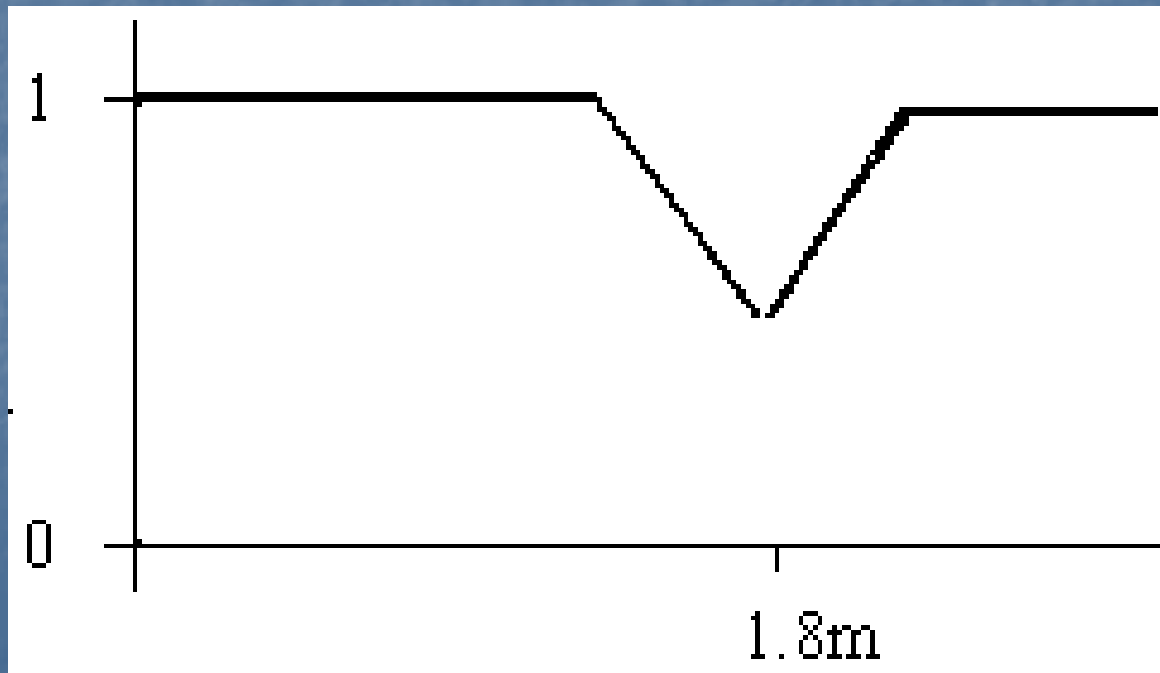


Fuzzy Set Operators

- Fuzzy Set:
 - Union
 - Intersection
 - Complement
- Many possible definitions
 - we introduce one possibility

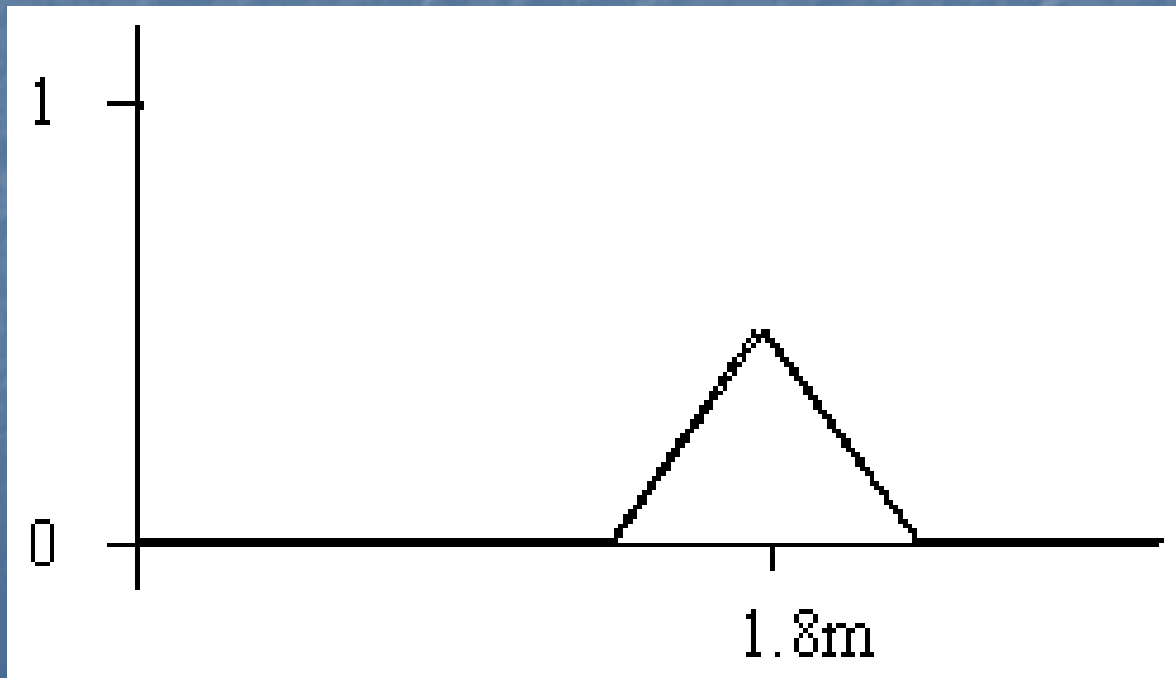
Fuzzy Set Union

- Union ($f_A(x)$ and $f_B(x)$) = $\max (f_A(x) , f_B(x))$
- Union (Tall(x) and Short(x))



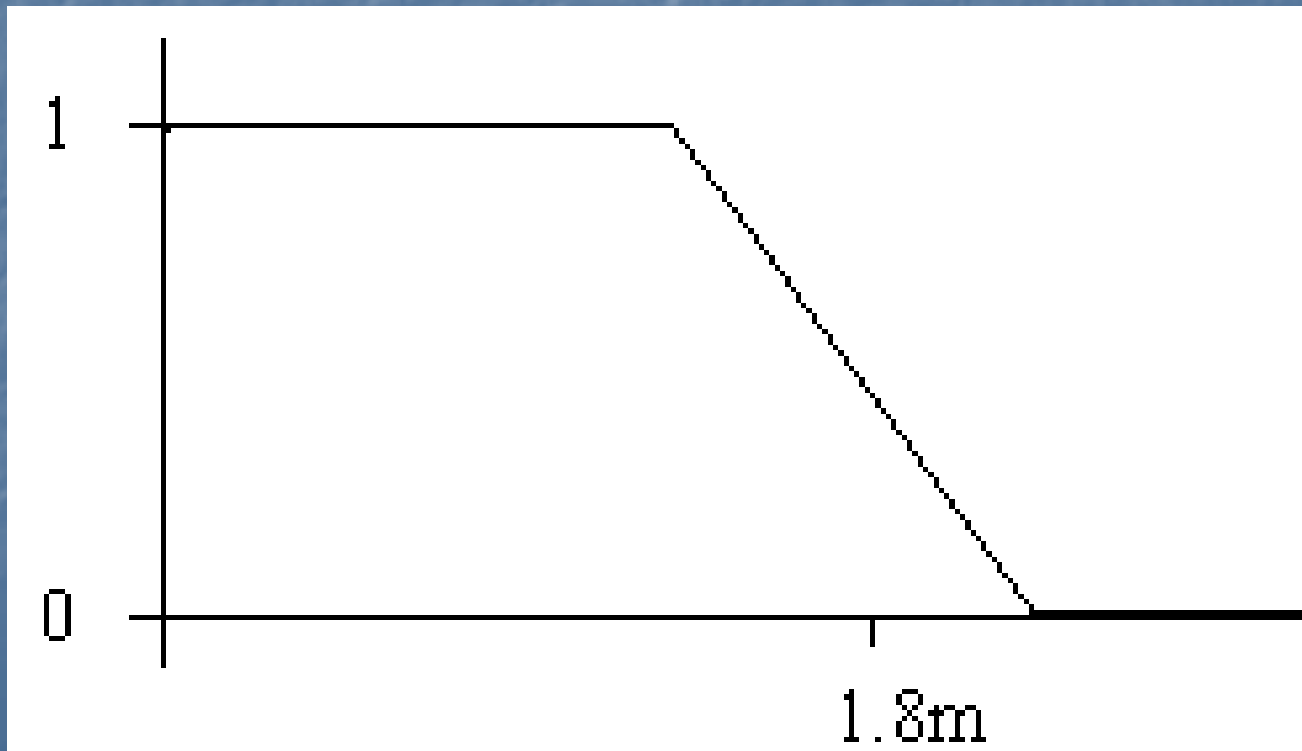
Fuzzy Set Intersection

- Intersection ($f_A(x)$ and $f_B(x)$) = $\min (f_A(x), f_B(x))$
- Intersection (Tall(x) and Short(x))



Fuzzy Set Complement

- $\text{Complement}(f_A(x)) = 1 - f_A(x)$
 - $\text{Not}(\text{Tall}(x))$



Fuzzy Logic Operators

- Fuzzy Logic:

- $\text{NOT } (A) = 1 - A$
- $A \text{ AND } B = \min (A, B)$
- $A \text{ OR } B = \max (A, B)$

Fuzzy Logic NOT

A	NOT A
0	1
0.25	0.75
0.5	0.5
0.75	0.25
1	0

Fuzzy Logic AND

A AND B B A	0	0.25	0.5	0.75	1.0
0	0	0	0	0	0
0.25	0	0.25	0.25	0.25	0.25
0.5	0	0.25	0.5	0.5	0.5
0.75	0	0.25	0.5	0.75	0.75
1	0	0.25	0.5	0.75	1

Fuzzy Logic OR

A OR B B A	0	0.25	0.5	0.75	1.0
0	0	0.25	0.5	0.75	1.0
0.25	0.25	0.25	0.5	0.75	1.0
0.5	0.5	0.5	0.5	0.75	1.0
0.75	0.75	0.75	0.75	0.75	1.0
1	1.0	1.0	1.0	1.0	1.0